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PRN-B2ME1120

To Interface a 16×2 LCD and Display Text Messages

```
#include <DHT.h>

#define DHTPIN 2

#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);

void setup() {

  Serial.begin(9600);

  dht.begin();

}

void loop() {

  float humidity = dht.readHumidity();

  float temperature = dht.readTemperature();

  // Check if reading failed

  if (isnan(humidity) || isnan(temperature)) {

    Serial.println("Failed to read from DHT11 sensor!");

    delay(2000);

    return;

  }

  Serial.print("Humidity: ");
```

```
  Serial.print(humidity);

  Serial.print(" %\t");

  Serial.print("Temperature: ");

  Serial.print(temperature);

  Serial.println(" °C");

  delay(2000);

}
```

CODE 2

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <DHT.h>

// DHT11 settings
#define DHTPIN 2
#define DHTTYPE DHT11

// Create DHT object
DHT dht(DHTPIN, DHTTYPE);

// Create LCD object
LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
  // Start Serial Monitor
  Serial.begin(9600);

  // Start DHT11
  dht.begin();

  // Start LCD
  lcd.init();
  lcd.backlight();
```

```
  // Startup message
  lcd.setCursor(0, 0);
  lcd.print("DHT11 Sensor");
  lcd.setCursor(0, 1);
  lcd.print("Starting...");

  delay(2000);
  lcd.clear();
}

void loop() {
  // Read DHT11
  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature();

  // Check whether the sensor reading is valid
  if (isnan(humidity) || isnan(temperature)) {
    Serial.println("DHT11 reading failed!");

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Sensor Error!");
```